

Answer key

3.6.48 Database Normalization: GATE DS&AI 2024 | Question: 36



Given the relational schema $R = (U, V, W, X, Y, Z)$ and the set of functional dependencies:

$$\{U \rightarrow V, U \rightarrow W, WX \rightarrow Y, WX \rightarrow Z, V \rightarrow X\}$$

Which of the following functional dependencies can be derived from the above set?

- A. $VW \rightarrow YZ$
- C. $VW \rightarrow U$

- B. $WX \rightarrow YZ$
- D. $VW \rightarrow Y$

gate-ds-ai-2024 databases database-normalization multiple-selects two-marks

Answer key

3.6.52 Database Normalization: GATE IT 2005 | Question: 70



In a schema with attributes A, B, C, D and E following set of functional dependencies are given

- $A \rightarrow B$
- $A \rightarrow C$
- $CD \rightarrow E$
- $B \rightarrow D$
- $E \rightarrow A$

Which of the following functional dependencies is NOT implied by the above set?

A. $CD \rightarrow AC$

B. $BD \rightarrow CD$

C. $BC \rightarrow CD$

D. $AC \rightarrow BC$

gateit-2005 databases database-normalization normal

Answer key 

3.6.52 Database Normalization: GATE IT 2005 | Question: 70



In a schema with attributes A, B, C, D and E following set of functional dependencies are given

- $A \rightarrow B$
- $A \rightarrow C$
- $CD \rightarrow E$
- $B \rightarrow D$
- $E \rightarrow A$

Which of the following functional dependencies is NOT implied by the above set?

- A. $CD \rightarrow AC$ B. $BD \rightarrow CD$ C. $BC \rightarrow CD$ D. $AC \rightarrow BC$

gateit-2005 databases database-normalization normal

Answer key

3.6.53 Database Normalization: GATE IT 2006 | Question: 60



Consider a relation R with five attributes V, W, X, Y , and Z . The following functional dependencies hold:
 $VY \rightarrow W, WX \rightarrow Z$, and $ZY \rightarrow V$.

Which of the following is a candidate key for R ?

- A. VXZ B. VXY C. $VWXY$ D. $VWXYZ$

gateit-2006 databases database-normalization normal

Answer key

3.6.40 Database Normalization: GATE CSE 2021 | Set 2 | Question: 40



Suppose the following functional dependencies hold on a relation U with attributes P, Q, R, S , and T :

- $P \rightarrow QR$
- $RS \rightarrow T$

Which of the following functional dependencies can be inferred from the above functional dependencies?

- A. $PS \rightarrow T$ B. $R \rightarrow T$ C. $P \rightarrow R$ D. $PS \rightarrow Q$

gatecse-2021-set2 multiple-selects databases database-normalization two-marks

Answer key

3.6.41 Database Normalization: GATE CSE 2022 | Question: 21



Consider a relation $R(A, B, C, D, E)$ with the following three functional dependencies.

$AB \rightarrow C; BC \rightarrow D; C \rightarrow E;$

The number of superkeys in the relation R is _____.

Answer key

3.6.35 Database Normalization: GATE CSE 2017 | Set 1 | Question: 16



The following functional dependencies hold true for the relational schema $R\{V, W, X, Y, Z\}$:

- $V \rightarrow W$
- $VW \rightarrow X$
- $Y \rightarrow VX$
- $Y \rightarrow Z$

Which of the following is irreducible equivalent for this set of functional dependencies?

A. $V \rightarrow W$

$V \rightarrow X$

$Y \rightarrow V$

$Y \rightarrow Z$

B. $V \rightarrow W$

$W \rightarrow X$

$Y \rightarrow V$

$Y \rightarrow Z$

C. $V \rightarrow W$

$V \rightarrow X$

$Y \rightarrow V$

$Y \rightarrow X$

$Y \rightarrow Z$

D. $V \rightarrow W$

$W \rightarrow X$

$Y \rightarrow V$

$Y \rightarrow X$

$Y \rightarrow Z$

Answer key

3.6.32 Database Normalization: GATE CSE 2015 | Set 3 | Question: 20



Consider the relation $X(P, Q, R, S, T, U)$ with the following set of functional dependencies

$$F = \{ \\ \{P, R\} \rightarrow \{S, T\}, \\ \{P, S, U\} \rightarrow \{Q, R\} \\ \}$$

Which of the following is the trivial functional dependency in F^+ , where F^+ is closure to F ?

- A. $\{P, R\} \rightarrow \{S, T\}$ B. $\{P, R\} \rightarrow \{R, T\}$
C. $\{P, S\} \rightarrow \{S\}$ D. $\{P, S, U\} \rightarrow \{Q\}$

gatecse-2015-set3 databases database-normalization easy

Answer key

3.6.33 Database Normalization: GATE CSE 2016 | Set 1 | Question: 21



Which of the following is NOT a superkey in a relational schema with attributes V, W, X, Y, Z and primary key VY ?

- A. $VXYZ$ B. $VWXZ$ C. $VWXY$ D. $VWXYZ$

gatecse-2016-set1 databases database-normalization easy

3.6.30 Database Normalization: GATE CSE 2014 | Set 1 | Question: 21



Consider the relation scheme $R = (E, F, G, H, I, J, K, L, M, N)$ and the set of functional dependencies

$$\{\{E, F\} \rightarrow \{G\}, \{F\} \rightarrow \{I, J\}, \{E, H\} \rightarrow \{K, L\}, \\ \{K\} \rightarrow \{M\}, \{L\} \rightarrow \{N\}\}$$

on R . What is the key for R ?

- A. $\{E, F\}$ B. $\{E, F, H\}$ C. $\{E, F, H, K, L\}$ D. $\{E\}$

gatecse-2012 databases easy database-normalization

Answer key

3.6.28 Database Normalization: GATE CSE 2013 | Question: 54



Relation R has eight attributes $ABCDEFGH$. Fields of R contain only atomic values. $F = \{CH \rightarrow G, A \rightarrow BC, B \rightarrow CFH, E \rightarrow A, F \rightarrow EG\}$ is a set of functional dependencies (FDs) so that F^+ is exactly the set of FDs that hold for R .

How many candidate keys does the relation R have?

A. 3

B. 4

C. 5

D. 6

gatecse-2013 databases database-normalization normal

Answer key

Answer key

3.6.24 Database Normalization: GATE CSE 2006 | Question: 70



The following functional dependencies are given:

$$AB \rightarrow CD, AF \rightarrow D, DE \rightarrow F, C \rightarrow G, F \rightarrow E, G \rightarrow A$$

Which one of the following options is false?

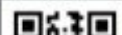
A. $\{CF\}^* = \{ACDEFG\}$
C. $\{AF\}^* = \{ACDEFG\}$

B. $\{BG\}^* = \{ABCDG\}$
D. $\{AB\}^* = \{ABCDG\}$

gatecse-2006 databases database-normalization normal

Answer key

3.6.25 Database Normalization: GATE CSE 2007 | Question: 62, UGCNET-June2014-II: 47



Answer key

3.6.20 Database Normalization: GATE CSE 2002 | Question: 2.25



From the following instance of a relation schema $R(A, B, C)$, we can conclude that:

A	B	C
1	1	1
1	1	0
2	3	2
2	3	2

- A. A functionally determines B and B functionally determines C
- B. A functionally determines B and B does not functionally determine C
- C. B does not functionally determine C
- D. A does not functionally determine B and B does not functionally determine C

Answer key 

3.6.15 Database Normalization: GATE CSE 2000 | Question: 2.24



Given the following relation instance.

X	Y	Z
1	4	2
1	5	3
1	6	3
3	2	2

Which of the following functional dependencies are satisfied by the instance?

A. $XY \rightarrow Z$ and $Z \rightarrow Y$

C. $YZ \rightarrow X$ and $X \rightarrow Z$

B. $YZ \rightarrow X$ and $Y \rightarrow Z$

D. $XZ \rightarrow Y$ and $Y \rightarrow X$

gatecse-2000 databases database-normalization easy

Answer key 

Answer key

3.6.13 Database Normalization: GATE CSE 1999 | Question: 1.24



Let $R = (A, B, C, D, E, F)$ be a relation scheme with the following dependencies

$C \rightarrow F, E \rightarrow A, EC \rightarrow D, A \rightarrow B$. Which one of the following is a key for R ?

A. CD

B. EC

C. AE

D. AC

gate1999 databases database-normalization easy

Answer key